

Fewer operations: no mean calculation
 RMSNorm: Fewer params: no bias term to store

original: $\text{FFN}(x) = xW_2 + b_2$

modern: $\text{FFN}(x) = G(xW_1)W_2$

Flops are not runtime

operator class	% Flop	% Runtime.
Tensor contraction	99.980	61.0
start. normalization	0.17	25.5
element-wise	0.13	13.5

RMS Norm rat still matter due to data movement.

RMS Norm: validation. w/ no RMS Norm.

Propply bin terms for. FFN network calculation.

#3 Activation: a zoo of activation functions. Gating activation
 ReLU, GeLU, ELU, ^{GeLU} GeLU, SeLU, SwiGLU, LiGLU. works ~~good~~ better.
 ↓ ↓
 GPT1/2. T5/Gemma2/3 palen/Llama/MiniSerial

#4: Serial VS Parallel layers: parallelize to Transformer
 ⇒ GPT J/PaLM/... do parallel layers Blocks

Architecture:

pre- vs post norm: ~~pre- vs~~ pre-norm work better

RMSnorm versus layernorm: RMSnorm is faster.

Activations: gated activation works more efficiently.

serial vs parallel layers:

Many variants in position embedding:

- Sine embeddings
- Absolute embeddings
- Relative embeddings
- RoPE embeddings: rotary position embeddings

AI mistakes / risks / harms / bias / misuses

Safeguards / anonymize privacy

proprietary

GW variants: PALM / Mistral / LLaMA / GPT

T5 model: 11B params $d_{\text{model}} = 1024$